

Software Engineering 3DX4

Dynamic Models and Control of Physical Systems

Winter 2025

INSTRUCTOR:

Prof. Mark Lawford

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Course Website: <https://avenue.mcmaster.ca/>

Office Hours: Wednesday 13:30-14:30 in MS Teams [Office Hours](#) channel or ITB/101b.

TAs:

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LECTURES:

- Location: BSB/B136
- Time: Tuesday, Wednesday, Friday 15:30-16:20.

Notes:

- This lecture schedule is based upon current university and public health guidelines and may be subject to changes during the term. Any changes to the schedule or course delivery will be communicated on the course announcements section on Avenue to Learn. Please check the announcements prior to attending class.
- Other information will sometimes be sent to your mcmaster.ca e-mail accounts and/or posted on the course website on Avenue and the MS Teams 3DX4 team. It is your responsibility to check your McMaster email account, the Avenue website and 3DX4 MS Team regularly.

LABS: Location: ITB/235. Time: See Mosaic for your scheduled section. Labs start the week of Jan. 12th.

See the 3DX4 website on Avenue for the 3DX4 Lab Safety Manual. You will be tested on the material in the manual and must successfully pass the quiz before you will be allowed to use the lab equipment.

DESCRIPTION:

Modeling of dynamic continuous physical phenomena in both continuous and discrete time. Control theory, stability analysis and feedback controller design. Application of computer control to continuous processes. Data analysis, empirical modeling.

MISSION:

To introduce the concepts of process dynamics and control as they apply to physical processes. To develop the necessary mathematical and analysis tools for process control. To understand the interaction between the dynamic behavior of processes and the design of controllers.

GRADING SCHEME:

- Assignment Quizzes 15%
- Labs 30%
- Midterm 20%
- Final Exam 35%

(The instructor reserves the right to conduct deferred examinations orally. All work is to be done individually unless you are explicitly directed otherwise.)

TEXT:

N.S. Nise, *Control Systems Engineering (8th Edition)*, John Wiley & Sons Inc., 2019. ISBN: 978-1-119-47422-7.

ADDITIONAL REFERENCES:

- Review previous math courses for linear algebra (matrices) and Laplace transforms as they will be used heavily in this course.
- Review material in Systems & Signals about mathematically modeling of physical systems.

DETAILED COURSE OUTLINE:

Introduction:

- Process Control Concepts
- Example Control Systems
- Basic Response Characteristics
- Open Loop vs. Closed Loop Control
- Analysis & Design Objectives
- The Design Process

Modeling in the Frequency Domain:

- Review of Laplace Transform
- The Transfer Function
- Obtaining Transfer Functions for specific application domains (e.g. Electrical, Mechanical, Chemical)

- Nonlinearities and Linearization
- Block Diagrams and Their Manipulation

Modeling in the Time Domain:

- State Space Models
- Transfer Function $\Rightarrow$ State Space Model
- State Space Model $\Rightarrow$ Transfer Function

Empirical Model Identification:

- Parameter identification
- Model validation

Time Response:

- Poles, Zeros and System Response
- First Order Systems
- Second Order Systems
- Higher Order Systems
- Stability
- Steady State Errors

Feedback Control Systems:

- Closed Loop Stability
- PID Controller
- Root Locus Analysis
- Root Locus Controller Design

Design via State Space:

- Controller Design and Controllability
- Observer Design and Observability

Digital Control Systems:

- Sampled Data Control Concepts
- Sampling and Quantization
- Aliasing and Prefiltering
- Discrete transfer functions and the Z transform
- Stability of discrete systems
- Steady State Error and Transient Response
- Sampled-data systems analysis and design

NOTES:

ACADEMIC INTEGRITY

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity.

It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: “Grade of F assigned for academic dishonesty”), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#), located at:

<https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/>

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one’s own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

AUTHENTICITY/PLAGIARISM DETECTION

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster’s use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

COURSES WITH AN ON-LINE ELEMENT

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

ONLINE PROCTORING

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

CONDUCT EXPECTATIONS

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

ACADEMIC ACCOMMODATION OF STUDENTS WITH DISABILITIES

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

REQUESTS FOR RELIEF FOR MISSED ACADEMIC TERM WORK

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar “Requests for Relief for Missed Academic Term Work”.

ACADEMIC ACCOMMODATION FOR RELIGIOUS, INDIGENOUS OR SPIRITUAL OBSERVANCES (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar’s Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

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The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

EXTREME CIRCUMSTANCES

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.

Learning Outcomes

Below we provide the post conditions for this course.

1. Students should know and understand
 - (a) the Laplace transform
 - (b) frequency domain analysis of control systems
 - (c) the concept of stability
 - (d) Routh-Hurwitz Criterion
 - (e) first and second order systems and their related transient response characteristics
 - (f) concept of steady state error
 - (g) root locus techniques
 - (h) what a PID controller is and how it works
2. Students should be able to
 - (a) model electrical, mechanical, and electromechanical systems in the frequency and time domain
 - (b) convert between transfer function and state-space representations of a system
 - (c) reduce multiple subsystems of a model to a single subsystem
 - (d) determine the transient response characteristics for first and second order systems
 - (e) determine the steady-state error of a system with respect to standard input functions
 - (f) analyze the stability of a system using roots/eigenvalues directly as well as using the Routh-Hurwitz Criterion
 - (g) design controllers in the frequency domain to satisfy stability, transient response, and steady state-error specifications
 - (h) sketch the root locus for transfer function of system
 - (i) use the root locus of a system to design a controller
 - (j) acquire data from a physical system and use it to perform system identification and validation
 - (k) Model discrete time linear systems using the Z transform.
 - (l) Perform basic analysis and design of digital control systems.
 - (m) Use state of the art tools such as Matlab, Labview and Maple to analyze, design and simulate linear feedback control systems.

1 Learning Outcome Rubrics

Topic	Below	Marginal	Meets	Exceeds
The Laplace transform	No knowledge of system Laplace Transform	Can complete some parts of Laplace and inverse Laplace transform, but not finish any problems	Performs basic steps of Laplace & inverse Laplace, but cannot apply beyond basic table lookup	Correctly applies all steps, able to use Laplace transform properties
Model electrical, mechanical, and electromechanical systems in the frequency and time domain	Unable to derive equations or solve for transfer function	Performs some of the steps to derive equations and solve for transfer functions	Demonstrates majority of knowledge required to derive equation TFs	(Near) Error-free derivation of all equations and TFs
The ability to evaluate and select appropriate modern tools	Unable to select and use tools	Uses some of the appropriate tools but not proficient enough to complete work	Uses appropriate tool effectively	Completes almost all aspects of work effectively using appropriate tools
Reduce multiple subsystems of a model to a single subsystem	Little to no ability to perform block diagram reduction	Can perform some simple block diagram reductions	Performs most block diagram reductions consistently	Performs all block diagram reductions correctly
Determine the transient response characteristics for first and second order systems	Little to no ability to determine transient response	Can perform some of the steps to compute transient response – knows to use Laplace and its inverse to start partial fractions expansion	Performs most steps correctly to compute transient response using Laplace transform	Performs all steps to correctly compute transient response using Laplace Transforms
Routh & Stability	Does not understand stability or how to evaluate it using techniques taught in class	Can perform some elements of stability analysis	Able to perform basic steps to determine stability	Can determine stability and number of poles in each half plane

Conversion between transfer functions and statespace representations	Does not understand how to convert between representations	Maybe understands formulas but cannot execute	Able to perform most of conversion	Easily converts between representations
Root Locus Design	Does not understand how to perform root locus and use it to design controller	Perform some elements of root locus design	Able to perform majority of design tasks using root locus	Able to perform root locus design to meet design specification
Steady state error	Does not understand how to determine steady state error	Can determine error transfer function or compute SS error but not design	Able to determine steady state error and design to meet SS error requirements	Able to design to meet design to meet SS error requirements and understands how theory relates to practice
Acquire data from a physical system and use it to perform system identification and validation	Unable to collect data and perform system ID	Able to collect data but not perform system ID	Performs relatively accurate system ID	Performs precise, well document system ID
Design controllers in the frequency domain to satisfy stability, transient response, and steady state-error specifications	Does not understand how to design a controller	Perform some elements of controller design	Able to perform majority of design tasks	Able to perform controller design to meet design specification
Model Discrete Time systems & perform analysis	Unable to create discrete time TF model	Able to partial model but not analyze	Models discrete system and understands some elements of analysis	Correctly models and analyzes discrete time systems

2 Relating Graduate Attribute Indicators with Learning Outcomes

1 (A knowledge base for engineering)

1.1 (Competence in Mathematics)

- The Laplace transform

1.3 (Competence in Engineering Fundamentals)

- Routh & Stability

1.4 (Competence in Specialized Engineering knowledge)

- Steady state error
- Conversion between transfer functions and statespace representations
- Reduce multiple subsystems of a model to a single subsystem
- Root Locus Design
- Model Discrete Time systems & perform analysis

2 (Problem Analysis)

2.1 (Demonstrates an ability to identify reasonable assumptions (including identification of uncertainties and imprecise information) that could or should be made before a solution path is proposed)

- Acquire data from a physical system and use it to perform system identification and validation
- Root Locus Design
- Design controllers in the frequency domain to satisfy stability, transient response, and steady state-error specifications

2.2 (Demonstrates an ability to identify a range of suitable engineering fundamentals (including mathematical techniques) that would be potentially useful for analyzing a technical problem)

- Routh & Stability
- Model Discrete Time systems & perform analysis

3 (Investigation)

3.3 (Estimates outcomes, uncertainties and determines appropriate data to collect)

- Determine the transient response characteristics for first and second order systems
- Acquire data from a physical system and use it to perform system identification and validation

4 (Design)

4.1 (Recognizes and follows an engineering design process.)

- Root Locus Design

- Steady state error

4.2 (Recognizes and follows engineering design principles including appropriate consideration of environmental, social and economic aspects as well as health and safety issues)

- Model electrical, mechanical, and electromechanical systems in the frequency and time domain
- Root Locus Design
- Design controllers in the frequency domain to satisfy stability, transient response, and steady state-error specifications
- Routh & Stability

4.3 (Proposes solutions to open-ended problems)

- Root Locus Design
- Design controllers in the frequency domain to satisfy stability, transient response, and steady state-error specifications

5 (Use of engineering tools)

5.1 (Evaluates and selects appropriate modern tools)

- The ability to evaluate and select appropriate modern tools

5.2 (Demonstrates an ability to use modern/state of the art tools)

not measured